PORTFOLIO ANALYSIS AND OPTIMIZATION

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Term Project Report
Trimester 7

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ABSTRACT

This report outlines a portfolio construction and optimization project using selected Indian stocks and a benchmark index. The aim is to determine an optimal capital allocation that maximizes expected returns while managing risk. Historical data from the Yahoo Finance API was cleaned to handle missing values and outliers, and key metrics such as returns, volatility, and Sharpe ratios were calculated. Random portfolio simulations were also performed to study the risk–return trade-off.

Using Modern Portfolio Theory and Markowitz mean–variance optimization, the study identified the Global Minimum Variance Portfolio (GMVP) and the Tangency Portfolio with the highest Sharpe ratio. The efficient frontier, Capital Market Line (CML), and Security Market Line (SML) were analyzed to evaluate portfolio efficiency and systematic risk. Results show that optimized portfolios outperform equal-weighted strategies and the benchmark, with the Tangency Portfolio offering the best risk-adjusted returns and the GMVP minimizing overall risk.

Index Terms— Portfolio Optimization, Modern Portfolio Theory, Efficient Frontier, Risk Analysis, Financial Data Science

1. INTRODUCTION

Portfolio optimization is a key part of investment management, aiming to combine assets to maximize expected returns while minimizing risk. This balance requires analyzing historical financial data and allocating capital strategically to reduce market volatility. Modern Portfolio Theory (MPT) provides a quantitative framework for constructing diversified portfolios along the efficient frontier, enabling a systematic understanding of the risk–return trade-off.

In this project, selected Indian stocks and a benchmark index are used to demonstrate data-driven portfolio construction. Historical price data obtained from the Yahoo Finance API were preprocessed to handle missing values and outliers. Key financial metrics, including returns, volatility, and Sharpe ratios, were computed to evaluate asset performance.

Furthermore, random portfolio simulations were conducted to explore the risk–return landscape and identify optimal portfolios. The Global Minimum Variance Portfolio (GMVP) and the Tangency Portfolio were derived using mean–variance optimization techniques. In addition, the Capital Market Line (CML) and Security Market Line (SML) were analyzed to assess systematic risk and portfolio efficiency.

2. PROBLEM STATEMENT AND OBJECTIVES

2.1. Problem Statement

Investors must allocate capital across different financial assets to optimize returns while controlling risk. Without a structured quantitative approach, portfolios can become poorly diversified, volatile, and inefficient. This project focuses on building an optimal equity portfolio by balancing expected return and risk using historical market data.

The study examines Indian stocks from multiple sectors along with the Nifty 50 index over a defined period. Portfolio construction follows practical constraints: non-negative asset weights, full capital allocation, and no short selling. The analysis assumes that historical price movements and correlations provide meaningful insights for portfolio optimization.

This problem remains highly relevant in modern investment decisions, as investors increasingly rely on quantitative methods for efficient capital allocation. It emphasizes the importance of diversification in reducing risk and shows that optimized portfolios can outperform simple or naive allocation strategies.

2.2. Objectives

The specific objectives of this project are as follows:

1. To collect and preprocess historical stock price and benchmark index data for portfolio analysis.
2. To compute key portfolio performance metrics, including expected return, volatility, and Sharpe ratio.
3. To analyze the impact of diversification through inter-asset correlations.
4. To simulate multiple portfolio combinations in order to examine the risk–return trade-off.
5. To design and implement optimized portfolios, including the Global Minimum Variance Portfolio (GMVP) and the Tangency Portfolio corresponding to the maximum Sharpe ratio.
6. To evaluate and compare the performance of optimized portfolios against equal-weighted portfolios and the benchmark index.

3. METHODOLOGY / APPROACH

This section outlines the methodology used to build and assess optimized stock portfolios. The process follows a structured sequence

from data collection and preprocessing to portfolio construction, optimization, performance assessment, and visualization. All steps remain fully reproducible using consistent datasets, parameters, and tools.

3.1. Overall Workflow

The overall workflow of the project consists of the following stages:

Data Collection: Adjusted closing prices for Indian stocks and the Nifty 50 index were gathered using the Yahoo Finance API over five years.

Data Preprocessing: Raw data were cleaned to handle missing values and outliers for reliable analysis.

Return and Risk Estimation: Daily returns were calculated, and metrics such as mean returns and the variance–covariance matrix were computed to capture asset behavior and co-movement.

Portfolio Simulation: Portfolios were generated by varying asset-weight combinations under realistic constraints to study the risk–return landscape.

Portfolio Optimization: Techniques identified portfolios, including the Global Minimum Variance Portfolio (GMVP) and the Tangency Portfolio with the highest Sharpe ratio.

Performance Evaluation and Visualization: Optimized portfolios were compared with equal-weighted portfolios and benchmarks using risk–return metrics and visualization tools such as the efficient frontier, CML, and SML.

Overall, the workflow moves from raw data to insights on portfolio risk and performance.

3.2. Technical Details

3.2.1. Data and Tools

Historical price data were retrieved via the Yahoo Finance API. The analysis used Python with NumPy, Pandas, SciPy, Matplotlib, and Seaborn, while asset choices and analysis duration were managed through a flexible, reproducible (`config.yaml`) file.

3.2.2. Return and Risk Computation

Asset returns were calculated as logarithmic daily returns. Expected returns were estimated as the mean of historical returns, while portfolio risk was quantified using the variance–covariance matrix of asset returns.

3.2.3. Portfolio Performance Metrics

For a given portfolio weight vector w, the expected portfolio return R_p and volatility σ_p were computed as

$$R_p = w^\top \mu, \tag{1}$$

$$\sigma_p = \sqrt{w^\top \Sigma w}, \tag{2}$$

where μ denotes the vector of expected asset returns and Σ represents the covariance matrix.

Risk-adjusted performance was measured using the Sharpe ratio, defined as

$$S = \frac{R_p - R_f}{\sigma_p}, \tag{3}$$

where R_f denotes the risk-free rate.

3.2.4. Optimization Method

Portfolio optimization was carried out using numerical optimization techniques under the following constraints:

- The sum of portfolio weights equals one.
- All portfolio weights are non-negative, prohibiting short selling.

The Global Minimum Variance Portfolio (GMVP) was obtained by minimizing portfolio variance, while the Tangency Portfolio was identified by maximizing the Sharpe ratio.

4. EXPERIMENTS AND RESULTS

Outlines the experimental setup and presents the results objectively.

4.1. Experimental Setup

The experiments were conducted using historical stock price data for five Indian companies—HCL Technologies, Axis Bank, Reliance Industries, Bharti Airtel, and Sun Pharma—along with the Nifty 50 index as a benchmark. The dataset spans five years of daily adjusted closing prices, yielding over 1,200 observations per asset. Data was obtained via the Yahoo Finance API and preprocessed by forward-filling missing values and removing outliers using Z-score filtering. Daily logarithmic returns were computed to estimate annualized returns and the variance–covariance matrix.

All experiments were implemented in Python using NumPy, Pandas, SciPy, Matplotlib, and Seaborn. Portfolio optimization was performed using the Sequential Least Squares Programming (SLSQP) algorithm, with a risk-free rate of 4% for Sharpe ratio computation. Performance was evaluated using annualized return, volatility, and Sharpe ratio.

4.2. Results

Table 1 summarizes the expected annual return and volatility of individual stocks.

Table 1. Expected Annual Return and Volatility of Individual Stocks

Ticker	Expected Annual Return	Annual Volatility
BHARTIARTL.NS	0.260951	0.223657
SUNPHARMA.NS	0.225563	0.206967
HCLTECH.NS	0.160695	0.237393
AXISBANK.NS	0.150084	0.248419
RELIANCE.NS	0.096513	0.225106

The correlation structure among asset returns is shown in Fig. 1.

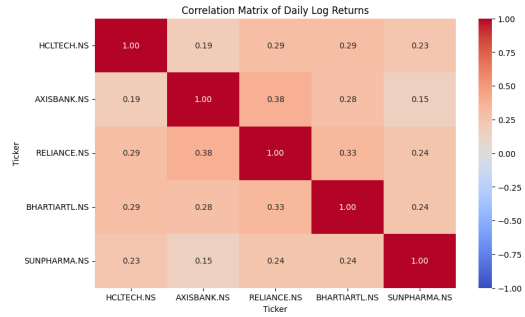

Fig. 1. Correlation matrix of daily log returns

The simulated portfolios and efficient frontier are illustrated in Fig. 2.

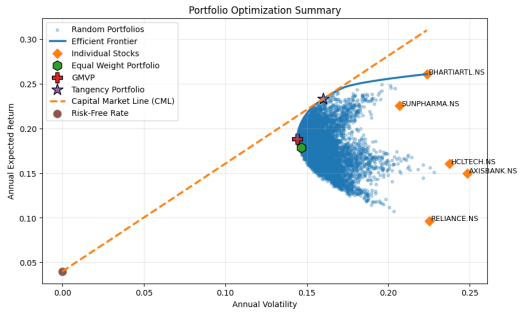

Fig. 2. Simulated portfolios with efficient frontier and Capital Market Line

Table 2 presents portfolio allocations and performance.

Table 2. Portfolio Weights and Performance Metrics

Ticker	Weight	Allocation (%)
Equal Weight Portfolio		
HCLTECH.NS	0.2	20.0
AXISBANK.NS	0.2	20.0
RELIANCE.NS	0.2	20.0
BHARTIARTL.NS	0.2	20.0
SUNPHARMA.NS	0.2	20.0
Return: 0.1788	Volatility: 0.1466	Sharpe: 0.9467
GMVP		
SUNPHARMA.NS	0.3170	31.7030
BHARTIARTL.NS	0.1888	18.8766
HCLTECH.NS	0.1761	17.6130
AXISBANK.NS	0.1638	16.3821
RELIANCE.NS	0.1543	15.4253
Return: 0.1885	Volatility: 0.1442	Sharpe: 1.0303
Tangency Portfolio		
BHARTIARTL.NS	0.4529	45.2871
SUNPHARMA.NS	0.4287	42.8724
HCLTECH.NS	0.0603	6.0299
AXISBANK.NS	0.0581	5.8105
RELIANCE.NS	0.0000	0.0000
Return: 0.2333	Volatility: 0.1601	Sharpe: 1.2075

Table 3 compares portfolios with the benchmark.

Table 3. Comparison of Portfolios with Benchmark

	Expected Return	Volatility	Sharpe Ratio
Benchmark	0.1199	0.1391	0.5746
Eq. Weight	0.1788	0.1466	0.9467
GMVP	0.1885	0.1442	1.0303
Tangency	0.2333	0.1601	1.2075

Fig. 3 shows investment growth over time.

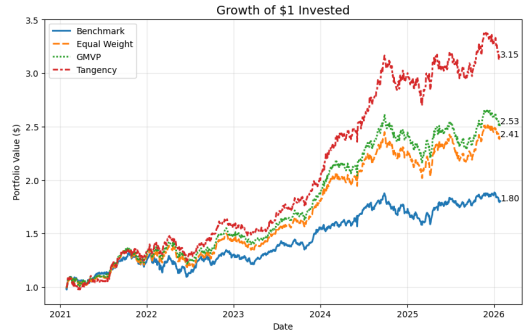

Fig. 3. Growth of one unit of investment

The Security Market Line is shown in Fig. 4.

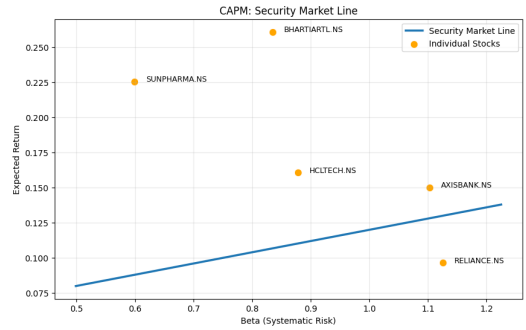

Fig. 4. Security Market Line (SML)

Table 4 lists the top five stocks by beta.

Table 4. Top Five Stocks by Beta

Ticker	Beta	Expected Return
RELIANCE.NS	1.1253	0.0965
AXISBANK.NS	1.1026	0.1501
HCLTECH.NS	0.8780	0.1607
BHARTIARTL.NS	0.8344	0.2610
SUNPHARMA.NS	0.5986	0.2256

5. DISCUSSION

This section summarizes and interprets the results, linking them to the project objectives while highlighting key strengths, trade-offs, and limitations of the portfolio optimization approach.

5.1. Performance of Different Portfolios

The results in Table 1 show noticeable differences in expected returns and volatility across individual stocks. BHARTIARTL.NS and SUNPHARMA.NS exhibit relatively high returns, while AXISBANK.NS and HCLTECH.NS show higher volatility, indicating greater risk exposure. These variations emphasize the importance of diversification and justify the need for portfolio-level optimization rather than relying on individual assets.

A clear performance gap exists between the benchmark, equal-weight portfolio, and optimized portfolios. The benchmark has the lowest Sharpe ratio, indicating weaker risk-adjusted performance (Table 3). This suggests that passive market exposure is less efficient than systematic portfolio construction using historical data and

quantitative methods.

The equal-weight portfolio outperforms the benchmark, highlighting the benefits of diversification by distributing capital evenly across assets. However, it remains less efficient than optimized portfolios (Table 3), demonstrating that equal allocation does not fully exploit return and risk characteristics of individual stocks.

The Global Minimum Variance Portfolio (GMVP) achieves the lowest volatility (Table 3) by selecting assets with favorable correlations. The correlation matrix confirms that the stocks are not perfectly correlated, which helps reduce overall portfolio risk (Figure 1). The GMVP slightly improves returns over the equal-weight portfolio while maintaining low risk, making it particularly suitable for risk-averse investors seeking stability.

The Tangency Portfolio delivers the highest Sharpe ratio and expected return (Table 3). Positioned on the Capital Market Line, it represents the optimal risk–return trade-off for the given risk-free rate (Figure 2). Its allocation favors assets with stronger return-to-risk profiles, leading to concentrated weights in certain stocks while others receive minimal or zero allocation (Table 2).

5.2. Risk-Free Rate, Capital Market Line, and Efficient Frontier

The inclusion of a risk-free rate enables the construction of the Capital Market Line (CML), which illustrates how investors can combine a risk-free asset with the tangency portfolio to achieve improved returns for a given level of risk. Portfolios lying below the CML are inefficient, while the tangency portfolio represents the optimal risky portfolio (Figure 2).

The efficient frontier represents portfolios that offer the maximum expected return for each level of volatility. Portfolios below this frontier are dominated by more efficient alternatives. Random portfolio simulations help visualize the entire risk–return space and confirm that optimized portfolios lie on or near the frontier, validating the effectiveness of the optimization process (Figure 2).

5.3. Security Market Line and Beta Analysis

The Security Market Line (SML) explains the relationship between systematic risk (beta) and expected return (Figure 4). Stocks with higher beta are expected to provide higher returns as compensation for increased market risk. RELIANCE.NS and AXISBANK.NS show higher sensitivity to market movements, while SUNPHARMA.NS exhibits relatively lower beta values (Table 4).

However, actual returns do not always align perfectly with the SML. These deviations may arise due to estimation errors, limited data samples, or market inefficiencies. This highlights a limitation of relying solely on CAPM-based assumptions and historical estimates.

5.4. Trade-Offs Observed

A clear risk–return trade-off is observed, where portfolios with higher expected returns also exhibit greater volatility. The tangency portfolio provides strong risk-adjusted performance but introduces concentration risk. In contrast, the GMVP offers stability with comparatively lower returns. Constraints such as no short selling and full investment improve practicality but restrict the solution space.

5.5. Limitations of the Approach

The study relies on historical data and assumes past trends will continue into the future. It does not explicitly account for changing market conditions, macroeconomic factors, or unexpected events. Transaction costs, taxes, and liquidity constraints are also ignored, which may impact real-world applicability. Additionally, expected returns and covariances are estimated from a limited dataset, introducing potential estimation risk.

5.6. Connection to Objectives

All objectives were successfully achieved, including data collection, preprocessing, metric computation, diversification analysis, and portfolio optimization. The comparison with the benchmark confirms that systematic portfolio optimization improves investment outcomes. Overall, the results demonstrate the practical value of quantitative portfolio analysis and provide a strong foundation for further enhancements and real-world applications.

6. CONCLUSION AND FUTURE WORK

This project examined and optimized a stock portfolio using historical market data and quantitative techniques. It demonstrated how risk and return can be measured, compared, and improved through systematic portfolio construction rather than relying on individual stocks or passive market exposure.

A major outcome was the comparison of different portfolio strategies. Diversification alone outperformed the benchmark, while optimized portfolios offered even stronger risk-adjusted returns. The Global Minimum Variance Portfolio reduced overall risk, making it suitable for conservative investors. In contrast, the Tangency Portfolio achieved the highest Sharpe ratio, showing superior performance when paired with a risk-free asset. Tools such as the efficient frontier, Capital Market Line, and Security Market Line clarified the trade-offs between risk, return, and market exposure.

The project also strengthened practical skills in data preprocessing, financial metric calculation, optimization, and interpretation. It highlighted the importance of quantitative methods in informed investment decisions and revealed the limitations of depending solely on historical data and simple assumptions.

Future work may expand the stock universe, analyze different market periods or benchmarks, and incorporate transaction costs, short selling, dynamic rebalancing, or alternative risk measures. Advanced approaches such as multi-factor models or real-time portfolio updates could further improve real-world applicability.

7. ARTIFACTS AND DEMONSTRATIONS

This section provides external resources supporting the implementation and results, ensuring reproducibility and transparency.

7.1. Source Code Repository

The complete source code for data ingestion, preprocessing, portfolio analysis, optimization, and visualization is available in a public repository. It includes well-documented Python scripts, configuration files, and Jupyter notebooks used to generate all results and figures.

Link: https://github.com/m1ttal/Portfolio-Analysis-and-Optimization.git

7.2. Video Demonstration

A short video walkthrough illustrating the project structure, execution steps, and key outputs is provided.

Link: https://youtu.be/njJfISGXK-M

8. REFERENCES

[1] Dr. Dipankar Mondal, *DA361 Financial Valuation and Portfolio Analytics*, Indian Institute of Technology Guwahati (IITG), Course Material, 2025.

[2] Francis, J., & Kim, K., *Modern Portfolio Theory: Foundations, Analysis, and New Developments*, Wiley Finance, Hoboken, NJ, 2013.

[3] Python Software Foundation, *Python Language Reference, Version 3.12*, 2023. Available: `https://www.python.org/`

[4] Ran, S., *yfinance: Yahoo! Finance market data downloader*, Python Package, 2019. Available: `https://pypi.org/project/yfinance/`

[5] *Jupyter Notebooks – a publishing format for reproducible computational workflows*, Positioning and Power in Academic Publishing: Players, Agents and Agendas, 2016.